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RF CONNECTOR AND COMMUNICATION DEVICE COMPRISING SAME

Abstract

Disclosed are an RF connector and a communication device comprising same. According to an embodiment of the present disclosure, a communication device is provided, the communication device comprising: an instrument box comprising a substrate mounting surface and a connection groove that is formed on the substrate mounting surface; a plurality of printed circuit boards arranged on the substrate mounting surface; a connector that is arranged in the connection groove and electrically connects the plurality of printed circuit boards; and a gasket arranged to cover at least part of the connector.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of International Application No. PCT/KR2023/018292, filed on Nov. 14, 2023, which claims priority to Korean Patent Application No. 10-2022-0155854, filed on Nov. 18, 2022, the disclosures of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present invention relates to an RF connector and a communication device including the same.

BACKGROUND ART

[0003] The description described in this part merely provides background information for the present disclosure, but does not constitute the related art.

[0004] In order to transmit and receive an radio frequency (RF) signal between printed circuit boards (PCBs), an RF connector is generally used. RF connectors include a single connector that carries a single frequency signal and a multi-connector that carries multiple signals in different frequency bands. In particular, in recent years, multi-connectors have been widely used to process signals in various frequency bands.

[0005] However, a connector for transmitting a frequency signal occupies a large volume, and in order to connect various printed circuit boards due to the structure and function of the connector, there is a problem in that a height difference between the printed circuit boards occurs, and the size of the device as a whole increases.

[0006] In addition, there is a limit to the range of frequency bands that may be processed by the connector, and there is a problem in that the unit cost of the component increases as a whole in order for the connector to transmit signals in many or a wide range of frequency bands.

CONTENTS OF THE INVENTION

Problem to be Solved

[0007] An RF connector and a communication device including the same according to the present disclosure may be configured to carry RF signals on the same plane or at similar heights to reduce the overall size of the device.

[0008] The RF connector and the communication device including the same according to the present disclosure may process signals in various frequency bands, and at the same time, may improve economic efficiency by lowering a unit price of a component.

[0009] The RF connector and the communication device including the same according to the present disclosure may be configured to have a compact size while having excellent radiating fin performance.

[0010] The RF connector and the communication device including the same according to the present disclosure may significantly improve the shielding effect by including a gasket disposed to cover the RF connector.

Method for Solving Problem

[0011] According to an embodiment of the present disclosure, there is provided a communication

device, including: an instrument box including a substrate mounting surface and a connection groove formed in the substrate mounting surface; a plurality of printed circuit boards disposed on the substrate mounting surface; a connector disposed in the connection groove and electrically connecting the plurality of printed circuit boards; and a gasket disposed to cover at least a part of the connector.

[0012] According to another embodiment of the present disclosure, there is provided an RF connector, including: an elastic connecting member, at least a part of which is elastically deformable, configured to electrically connect the printed circuit boards; a support member for supporting the elastic connecting member; and a gasket disposed to cover at least a part of the elastic connecting member.

Effects of the Invention

[0013] According to the present disclosure, an RF connector and a communication device including the same are configured to transmit RF signals on the same plane or at a similar height, which has the effect of reducing the size of the entire device.

[0014] According to the present disclosure, the RF connector and the communication device including the same have an effect of processing signals in various frequency bands, and improving economic efficiency by lowering a unit cost of a component.

[0015] According to the present disclosure, the RF connector and the communication device including the same have an effect of being configured to have a compact size while having excellent radiating fin performance.

[0016] According to the present disclosure, the RF connector and the communication device including the same include a gasket disposed so as to cover the RF connector, which has the effect of significantly improving the shielding effect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is an exploded perspective view of some configurations of a communication device according to an embodiment of the present disclosure.

[0018] FIG. 2A is a cross-sectional view of a portion of a communication device according to an embodiment of the present disclosure, showing when printed circuit boards are not disposed.

[0019] FIG. 2B is a cross-sectional view of a portion of a communication device according to an embodiment of the present disclosure, showing when printed circuit boards are disposed.

[0020] FIG. 3 is a perspective view of a portion of a communication device according to an embodiment of the present disclosure.

[0021] FIG. 4 is a perspective view separately illustrating only a connector of a communication device according to an embodiment of the present disclosure.

[0022] FIG. 5 is a diagram illustrating examples of a connector that is integrally formed via an insert injection method according to an embodiment of the present disclosure.

[0023] FIG. 6A is a cross-sectional view of a portion of a communication device according to another embodiment of the present disclosure, showing when printed circuit boards on which a printed board connected with a connector has the same height are not disposed.

[0024] FIG. 6B is a cross-sectional view of a portion of a communication device according to another embodiment of the present disclosure, showing when printed circuit boards on which a printed board connected with a connector has the same height are disposed.

[0025] FIG. 7 is a perspective view of a portion of a communication device according to another embodiment of the present disclosure.

[0026] FIG. 8 is a perspective view separately illustrating only a connector of a communication device according to another embodiment of the present disclosure.

[0027] FIG. **9** is a perspective view of a portion of a device configured for simulating connector performance of a communication device according to the present disclosure.

[0028] FIG. **10** is a perspective cross-sectional view of FIG. **9**.

[0029] FIG. **11** is a simulation result for the device configuration of FIGS. **9** and **10**.

[0030] FIG. **12A** is a cross-sectional view of a portion of a communication device according to another embodiment of the present disclosure, showing when printed circuit boards are not disposed.

[0031] FIG. **12B** is a cross-sectional view of a portion of a communication device according to another embodiment of the present disclosure, showing when printed circuit boards are disposed.

[0032] FIG. **13** is a perspective view of a portion of a communication device according to another embodiment of the present disclosure.

[0033] FIG. **14** is an exploded perspective view of a connector according to another embodiment of the present disclosure.

[0034] FIG. **15** is a cross-sectional view of a gasket according to another embodiment of the present disclosure.

[0035] FIG. **16** is a simulation result for another embodiment of the present disclosure.

[0036] FIG. **17** is an exploded perspective view of a connector including a gasket without an insertion groove according to another embodiment of the present disclosure.

[0037] FIG. **18** is an exploded perspective view of a connector including a gasket with an insertion groove according to another embodiment of the present disclosure.

[0038] FIG. **19** is a simulation result for the connector shown in FIGS. **17** and **18**.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0039] Hereinafter, some embodiments of the present disclosure will be described in detail with reference to exemplary drawings. Note that when components in each drawing are denoted by reference numerals, the same components are denoted by the same numerals as much as possible even if they are denoted on different drawings. In addition, in describing the present disclosure, if it is determined that a specific description of a related known configuration or function may obscure the subject matter of the present disclosure, the detailed description thereof will be omitted.

[0040] In describing the components of the present invention, the terms “first,” “second,” “A,” “B,” “(a),” “(b),” and the like may be used. These terms are only used to distinguish the components from other components, and the nature, sequence, order, or the like of the components is not limited by these terms.

[0041] Where an element is described as being “connected,” “coupled” or “accessed” to another element, it will be understood that the element may be directly connected or connected to the other element, but that another element may also be “connected,” “coupled” or “accessed” between each element.

[0042] Throughout the specification, when a part “includes” or “has” an element, unless otherwise specified, the part may further include other elements instead of excluding the other elements.

[0043] The terms “. . . unit,” “module” and the like described in the specification mean a unit that processes at least one function or operation, and may be implemented by hardware or software, or a combination of hardware and software.

[0044] Unless otherwise specified, it should be understood that the description of one embodiment may be applied to other embodiments.

[0045] The description set forth below in connection with the appended drawings is intended to describe exemplary embodiments of the invention and is not intended to represent the only embodiments in which the invention may be practiced.

[0046] FIG. **1** is an exploded perspective view of some configurations of a communication device according to an embodiment of the present disclosure.

[0047] FIG. **2A** is a cross-sectional view of a portion of a communication device according to an embodiment of the present disclosure, showing when printed circuit boards are not disposed.

[0048] FIG. 2B is a cross-sectional view of a portion of a communication device according to an embodiment of the present disclosure, showing when printed circuit boards are disposed.

[0049] FIG. 3 is a perspective view of a portion of a communication device according to an embodiment of the present disclosure.

[0050] Referring to FIG. 1 to FIG. 3, a communication device according to an embodiment of the present disclosure includes all or a part of an instrument box **20**, a first printed circuit board **10**, a second printed circuit board **12**, and one or more connectors **300**.

[0051] The instrument box **20** includes a substrate mounting surface **210** on which the first printed circuit board **10** and the second printed circuit board **12** are mounted, and one or more connection grooves **220** formed in the substrate mounting surface **210**. The instrument box **20** may be formed of a thermally conductive rigid member, for example, an aluminum alloy or a stainless steel alloy.

[0052] In one embodiment of the invention, the instrument box **20** is configured to form the exterior of the communication device and support the device structure, and is illustrated as being an outer housing configured to receive printed circuit boards. The outer housing includes a plurality of radiating fins **21** formed on at least some surfaces thereof.

[0053] However, the present invention is not limited thereto, and the instrument box **20** may be a separate component other than the outer housing, for example, may be a separate thermally conductive rigid member that supports the printed circuit boards between the outer housing and the printed circuit board.

[0054] The substrate mounting surface **210** may include a first mounting surface **212** on which the first printed circuit board **10** is mounted and a second mounting surface **214** on which the second printed circuit board **12** is mounted. The first mounting surface **212** and the second mounting surface **214** may be formed to have a step with respect to each other.

[0055] In one embodiment of the present invention, the first printed circuit board **10** and the second printed circuit board **12** may have different thicknesses. The size of the step between the first mounting surface **212** and the second mounting surface **214** may correspond to a difference in thickness between the first printed circuit board **10** and the second printed circuit board **12**. As a result, the upper surface of the first printed circuit board **10** (the surface opposite to the surface supported by the substrate mounting surface **210**) and the upper surface of second printed circuit board **12** (the surface opposite the surface supported by substrate mounting surface **210**) may be disposed on the same plane or on a plane equivalent thereto.

[0056] Although not shown here, the coplanar arrangement of the first printed circuit board **10** and the second printed circuit board **12** may serve as an advantage in reducing or optimizing the overall size and weight of the communication device, in arranging any connectors **300** that directly connect the two, or filters or other elements that support the two or are installed on the top surface.

[0057] The one or more connection grooves **220** are formed in the substrate mounting surface **210**, and for example, a plurality of connection grooves **220** may be formed along a boundary between the first mounting surface **212** and the second mounting surface **214**.

[0058] The first printed circuit board **10** may include a plurality of devices mounted thereon. In the RF connector according to an embodiment of the present disclosure, the first printed circuit board **10** may be an amplifier (Amp) board, but is not limited thereto.

[0059] The second printed circuit board **12** may include a plurality of devices disposed adjacent to and spaced apart from the first printed circuit board **10** on one side of the first printed circuit substrate **10** and mounted thereon. In the RF connector according to an embodiment of the present disclosure, the second printed circuit board **12** may be, but is not limited to, a digital board.

[0060] The one or more connectors **300** are disposed in the connection groove **220** and configured to electrically connect the first printed circuit board **10** and the second printed circuit board **12**.

[0061] FIG. 4 is a perspective view separately illustrating only a connector of a communication device according to an embodiment of the present disclosure.

[0062] Referring to FIG. 4, the connector **300** may include a support member **310** and an elastic

connecting member **320**.

[0063] The support member **310** may contact at least a part of the instrument box **20** in the connection groove **220**. The support member **310** is an insulating material, and may be made of a high heat-resistant thermoplastic, for example, polycarbonate, Lusep, or Teflon.

[0064] The elastic connecting member **320** may be made of an electrically conductive material, and may be made of a material having elasticity and suitable for transmitting a high frequency signal (RF signal), for example, beryllium copper (BeCu) or stainless steel alloy (SUS). The elastic connecting member **320** is disposed between the support member **310** and the first printed circuit board **10** and/or the second printed circuit board **12**, and is configured to electrically connect the first printed circuit substrate **10** and the second printed circuit substrate **12**.

[0065] In addition, the elastic connecting member **320** may include a plate-shaped base **324**, a first side wall **325** formed at one end of the plate-shaped base **324**, a first contact portion **326** formed at the one end of the first side wall **326**, a second side wall **327** formed at the other end of the plate-shaped base **324** and a second contact portion **328** formed at the one ends of the second side wall **326**. The elastic connecting member **320** may be formed by bending processing.

[0066] As described above, the substrate mounting surface **210** may have a step corresponding to a difference in thickness between the first printed circuit board **10** and the second printed circuit board **12**. In addition, lengths (or vertical heights) of the first side wall **325** and the second side wall **327** of the elastic connecting member **320** may be different to correspond to a size of a step of the substrate mounting surface **210**.

[0067] The support member **310** may include a support base **312**, a protruding mount portion **314** protruding from one surface of the support base **312** and configured to mount the elastic connecting member **320**, and a fixing member **316** formed to fix the elastic connecting member **310** on the protruding mount portion **314**.

[0068] Referring back to FIG. 2A and FIG. 4, the plate-shaped base **324** of the elastic connecting member **320** is supported on the protruding mount portion **314**, so that a deformation gap **222** may be formed between the plate-shaped basis **324** of the elastic connecting member **320** and the support base **312** of the support member **310**.

[0069] That is, in an embodiment of the present disclosure, the connector **300** may include a deformation gap **222** formed between the support base **312** and the elastic connecting member **320**, so that when the first printed circuit board **10** and/or the second printed circuit board **12** are mounted on the substrate mounting surface **210**, the first contact portion **326** and the second contact portion **328** are respectively pressed by the lower surfaces of the printed circuit boards **10** and **12**, and the elastic connecting members **320** are elastically deformable (see FIG. 2B).

[0070] Further, in an embodiment of the present invention, the fixing member **316** may include a fixing member **316** formed on the protruding mount portion **314**. The elastic connecting member **320** may include a through hole into which the fixing member **316** is inserted.

[0071] FIG. 5 is a diagram illustrating examples of a connector that is integrally formed through an insert injection method according to an embodiment of the present disclosure.

[0072] Referring to FIG. 5, the connector **300** according to the present disclosure may be manufactured in an insert injection manner. That is, the elastic connecting member **320** and the support member **310** may be integrally formed through insert injection. However, the present disclosure is not limited thereto, and the elastic connecting member **320** and the support member **310** may be separately manufactured and integrated through a separate assembly process.

[0073] In one illustrated embodiment of the invention, the connector **300** is illustrated as having a structure and size for transmitting and receiving high frequency signals. However, the present invention is not limited thereto. The connector **300** may be used in variations to have a structure sized to carry signals other than high frequency signals, for example, a power source or an analog or digital low frequency signal.

[0074] FIG. 6A is a cross-sectional view of a portion of a communication device according to

another embodiment of the present disclosure, showing when printed circuit boards on which a printed board connected with a connector has the same height are not disposed.

[0075] FIG. **6B** is a cross-sectional view of a portion of a communication device according to another embodiment of the present disclosure, showing when printed circuit boards on which a printed board connected with a connector has the same height are disposed.

[0076] FIG. **7** is a perspective view of a portion of a communication device according to another embodiment of the present disclosure.

[0077] FIG. **8** is a perspective view separately illustrating only a connector of a communication device according to another embodiment of the present disclosure.

[0078] A communication device according to another embodiment of the present disclosure will be described with reference to FIGS. **6** to **8**. Description of overlapping parts with one embodiment of the present disclosure will be omitted, but it should be noted that features of one embodiment according to the present disclosure may be applied to other embodiments in the same or similar manner to the extent that they are not disposed in other embodiments of the present disclosure.

[0079] According to another embodiment of the present disclosure, the first mounting surface **212** and the second mounting surface **214** may be planar so as to have the same or similar height, i.e., so as not to have a step. The first printed circuit board **10** and the second printed circuit board **12** may be the same or similar in thickness. The upper side of the first printed circuit board **10** and the upper side of the second printed circuit board **12** may thus be arranged on the same plane or a similar plane.

[0080] When the first printed circuit board **10** and the second printed circuit board **12** are formed to have the same or similar thickness, the substrate mounting surface **210** may be formed to have a planar shape without a step. In this case, lengths (or vertical heights) of the first side wall **325** and the second side wall **327** of the elastic connecting member **320** may be the same or similar.

[0081] FIG. **9** is a perspective view of a portion of a device configured for simulating connector performance of a communication device according to the present disclosure.

[0082] FIG. **10** is a perspective cross-sectional view of FIG. **9**.

[0083] FIG. **11** is a simulation result for the device configuration of FIGS. **9** and **10**.

[0084] FIG. **10** shows a communication device according to an embodiment of the present disclosure, that is, the printed circuit boards **10** and **12** have different thicknesses, and the side walls **325** and **327** have different heights. However, it should be noted that the effects of the present disclosure described below may not only be achieved according to an embodiment, but may also be achieved in other embodiments of the present disclosure. That is, the thicknesses of the printed circuit board **10** and **12** may be the same or similar, and the heights of the side walls **325** and **327** may be the same, or similar.

[0085] Referring to FIG. **9** to FIG. **11**, a virtual digital board and a virtual amplifier board structure are used to simulate the connector performance of the communication device according to the present invention.

[0086] The virtual digital board (the second printed circuit board **12**) and the virtual amplifier board (the first printed circuit board **10**) each include a first electrode **54** and a second electrode **52** that are in contact with and electrically connected to the first contact portion **326** and the second contact portion **328** of the elastic connecting member **320**.

[0087] For the purpose of simulation, a plurality of via holes **56** were illustrated surrounding the periphery of the first electrode **54** and the second electrode **52**, and a port **58** for signal transmission was installed in each of the first electrode **52** and the second electrode **54**, respectively.

[0088] As with the simulation result shown in FIG. **11**, for this configuration, the return loss of the connector **300** was found to be 20 dB. In addition, it may be seen that when the interval of the structure of the connector **300** is 10 mm (1 cm), and when the frequency of the high-frequency signal is 2.16 GHz, the isolation value is 50 dB. It may also be seen that the insertion loss is at most 0.1 dB.

[0089] It may be seen that the isolation degree varies according to the interval of the connector structure, and the connector structure interval according to the connector structure and the high-frequency signal type may be determined and optimized by referring to the simulation result.

[0090] According to the present invention, the heat management of the first printed circuit board **10** and the second printed circuit board **12**, which are a kind of heat source, in particular, the radiating performance through direct heat conduction, may be an important factor in determining the performance of the entire device.

[0091] Thus, in one embodiment of the invention the first printed circuit board **10** and the second printed circuit board **12** are arranged in direct contact with the instrument box **20**, for example the outer housing.

[0092] However, the region in which the first printed circuit board **10** and the second printed circuit board **12** are spaced apart from each other has a relatively small influence on the radiating fin performance of the printed circuit boards due to direct conduction.

[0093] The present invention forms a connecting structure for connecting both substrates in a spaced-apart arrangement region between these printed circuit boards. Since the instrument box **20**, for example, the outer housing, is itself a metallic shielding material, the connector according to an embodiment of the present disclosure may maintain an effective shielding state with respect to the outside inside the connection groove **220**. That is, even if the connector is not covered with a separate RF shielding member, the connector of the communication device according to the present invention may have a sufficient shielding effect (degree of isolation) as revealed in the simulation results above.

[0094] Furthermore, the communication device having a compact size while having excellent radiating fin performance may be provided because as large an area as possible of the printed circuit boards may have a structure in direct surface contact with the instrument box **20**.

[0095] FIG. **12A** is a cross-sectional view of a portion of a communication device according to another embodiment of the present disclosure, showing when printed circuit boards are not disposed.

[0096] FIG. **12B** is a cross-sectional view of a portion of a communication device according to another embodiment of the present disclosure, showing when printed circuit boards are disposed.

[0097] FIG. **13** is a perspective view of a portion of a communication device according to another embodiment of the present disclosure.

[0098] FIG. **14** is an exploded perspective view of a connector according to another embodiment of the present disclosure.

[0099] FIG. **15** is a cross-sectional view of a gasket according to another embodiment of the present disclosure.

[0100] Before a communication device according to another embodiment of the present disclosure is described, overlapping parts with the communication devices according to the embodiments of the present disclosure described above will be omitted, but it should be noted that the contents (for example, the support base, the protruding mount portion, the fixing member, the plate-shaped base, the first side wall, the first contact portion, the second side wall, the second contact portion, and the like) of the communication device according to one embodiment and another embodiment of the disclosure may be applied equally or similarly to the communication device of another embodiment of the invention to the extent that they are not arranged.

[0101] Referring to FIGS. **12-15**, a communication device according to another embodiment of the present disclosure may include a connector **1400** and a gasket **1430**.

[0102] The connector **1400** may include a support member **1410** and an elastic connecting member **1420**.

[0103] The support member **1410** may include a support base **1412**, a protruding mount portion **1414**, a fixing member **1416**, and a protruding insertion portion **1418**.

[0104] The elastic connecting member **1420** may include a plate-shaped base **1424**, a first side wall

1425, a first contact portion **1426**, a second side wall **1427**, and a second contact portion **1428**.

[0105] The gasket **1430** may include a body **1432** and an insertion groove **1434**.

[0106] The support member **1410** according to another embodiment of the present disclosure may further include a protruding insertion portion **1418** to be coupled to the gasket **1430**. The protruding insertion portions **1418** may be formed on both sides of the protruding mount portion **1414**. The protruding insertion portion **1418** is formed to protrude from the support base **1412**. When the protruding insertion portion **1418** is inserted into the insertion groove **1434** of the gasket **1430**, the support member **1410** and the gasket **1330** may be coupled to each other. The elastic connecting member **1420** is disposed between the support member **1410** and the gasket **1430**. The protruding insertion portion **1418** may be formed in a shape corresponding to the shape of the insertion groove **1434** of the gasket **1430**, and the height of the protruding insertion portion **1418** may be the same as or similar to the height of the insertion groove **1434**, so that the support member **1410** and the gasket **1330** are firmly coupled. The protruding insertion portion **1418** may be formed to have a height higher than that of the protruding mounting portion **1414**.

[0107] Hereinafter, the gasket **1430** will be described as being separate from the connector **1400**, but it should be noted that the gasket **1430** and the connector **1400** do not necessarily have to be separate configurations, and the gasket **1430** may be understood as one configuration included in the connector **1400**.

[0108] The gasket **1430** is disposed to cover at least a part of the connector **1400** to improve a shielding effect of the connector **1400**. The gasket **1430** may be disposed to cover at least a portion of the elastic connecting member **1420** of the connector **1400**. The gasket **1430** may be disposed to cover at least a portion of the plate-shaped base **1424** of the elastic connecting member **1420**. As shown in FIGS. **13** and **14**, the gasket **1430** may be disposed to intersect the connector **1400**. Thus, the shielding effect of the connector **1400** may be improved while a plurality of printed circuit boards are electrically connected. The gasket **1430** may be disposed to be orthogonal to the connector **1400**.

[0109] The gasket **1430** may be coupled with the connector **1400**. The gasket **1430** may be configured to allow insertion of at least a portion of the connector **1400**, the protruding insertion portion **1418** of the support member **1410**. The gasket **1430** may include a body **1332** and an insertion groove **1434**. The protruding insertion portion **1418** of the support member **1410** may be inserted into the insertion groove **1434**. As described above, the insertion groove **1434** and the protruding insertion portion **1418** may be formed in a shape corresponding to each other, so that the gasket **1430** and the connector **1400** may be firmly coupled. The height of the insertion groove **1434** may be the same as or similar to the height of the protruding insertion portion **1418**.

[0110] Referring to FIG. **15**, the height h of the insertion groove **1434** may be equal to or greater than half of the total height H of the gasket **1430**. For example, when the total height H of the gasket **1430** is 1.2 mm, the height of the insertion groove **1434** may be 0.7 mm. Therefore, the gasket **1430** and the connector **1400** are firmly coupled without increasing the package size, and the shielding effect of the connector **1400** may also be improved.

[0111] The gasket **1430** may be made of aluminum to improve the shielding effect of the connector **1400**, but the material of the gasket **1430** is not limited thereto, and various materials for improving the shielding effect of the connector **1400** may be applied.

[0112] FIG. **16** is a simulation result for another embodiment of the present disclosure including a gasket **1430**.

[0113] Referring to FIG. **16**, the return loss of the connector **1400** of the communication device including the gasket **1430** is found to be 21.7451 dB when the frequency of the high frequency signal is 4 GHz. In addition, when the frequency of the high-frequency signal is 4 GHz with respect to S (**1,3**), the isolation value is 124.0433 dB, and it may be confirmed that the shielding effect is significantly improved compared to the case where the gasket **1430** is not included (see FIG. **11**). In addition, it may be seen that the insertion loss is 0.1711 dB when the frequency of the high

frequency signal is 4 GHz.

[0114] As described above, in the communication device including the gasket **1430** according to the present disclosure, the gaskets **1430** are disposed so as to cover at least a part of the connector **1400**, and thus there is an effect that the shielding effect for the connector **1400** is significantly enhanced.

[0115] FIG. **17** is an exploded perspective view of a connector including a gasket without an insertion groove according to another embodiment of the present disclosure.

[0116] Referring to FIG. **17**, in the gasket **1430** according to another embodiment of the present disclosure, the above-described insertion groove **1434** may not be formed, and both an upper surface and a lower surface may be formed to be flat. When the insertion groove **1434** is not formed in the gasket **1430**, the gasket **1430** may be mounted on the support member **1410**, for example, on the protruding mount portion **1414** (see FIG. **14**) or the protruding insertion portion **1418** (see FIG. **4**) to cover the connector **1400**. When the gasket **1430** does not include the insertion groove **1434** and is formed in a flat shape, the shielding effect of the connector **1400** is improved, and the structure is simple and the manufacturing cost is reduced.

[0117] FIG. **18** is an exploded perspective view of a connector including a slotted gasket according to another embodiment of the present disclosure.

[0118] Referring to FIG. **18**, an insertion groove **1434** may be formed in the gasket **1430** according to another embodiment of the present disclosure as described above. For example, an insertion groove **1434** may be formed on a lower surface of the gasket **1430**. As illustrated in FIG. **18**, one surface (e.g., a lower surface) of the gasket **1430** may be formed to have a step, and the insertion groove **1434** may be formed on the one surface formed to have a step. When a part of the support member **1410** (e.g., the protruding insertion portion **1418** (see FIG. **14**)) is inserted into the insertion groove **1434**, the connector **1400** and the gasket **1430** may be coupled to each other. When the insertion groove **1434** is formed in the gasket **1430** and the connector **1400** and the gasket **1430** are coupled by using the insertion groove, return loss may be improved as described below.

[0119] FIG. **19** is a simulation result for the connector shown in FIGS. **17** and **18**.

[0120] Referring to FIG. **19**, it is shown that the return loss A of the connector **1400** covered with the flat-shaped gasket **1430** in which the insertion groove **1434** described with reference to FIG. **17** is not formed is 19.1064 dB when the frequency of the high frequency signal is 4 GHz, and the return loss B of the connector **1400** covered with the stepped gasket **1430** including the insertion groove **1434** described with reference to FIGS. **18A** and **18B** is 26.7269 dB when the frequencies of the high frequency signals are 4 GHz. That is, it may be seen that the insertion groove **1434** is formed in the gasket **1430**, and when the connector **1400** and the gasket **1430** are coupled by using this, the return loss is improved. However, even when the gasket **1430** having a flat shape illustrated in FIG. **17** is used, a relatively excellent return loss may be expected while improving the shielding effect of the connector **1400**, and the structure is simple and the manufacturing cost is reduced, so that the gasket **1430** having an appropriate structure may be selected and applied as necessary.

[0121] The foregoing descriptions are merely illustrative of the technical concept of the present embodiment, and various modifications and variations may be made by those skilled in the art without departing from the essential characteristics of the present embodiment. Therefore, the present embodiments are not intended to limit the technical concept of the present embodiments, but are intended to be illustrative, and the scope of the technical concept of this embodiment is not limited by these embodiments. The protection scope of the present embodiment is to be construed according to the following claims, and all technical ideas within the scope equivalent thereto are construed as being included in the scope of rights of the present embodiment.

Description of Signs

Claims

1. A communication device, comprising: an instrument box including a substrate mounting surface and a connection groove formed in the substrate mounting surface; a plurality of printed circuit boards disposed on the substrate mounting surface; a connector disposed in the connection groove and electrically connecting the plurality of printed circuit boards; and a gasket disposed to cover at least a part of the connector.
2. The communication device of claim 1, wherein the gasket is configured such that at least a portion of the connector may be inserted.
3. The communication device of claim 2, wherein the gasket comprises: an insertion groove into which a body portion and at least a part of the connector are inserted.
4. The communication device of claim 3, wherein the connector comprises: a support member disposed in the connection groove and in contact with at least a part of the instrument box; and a conductive elastic connecting member disposed between the support member and the plurality of printed circuit boards, and the gasket is disposed so that at least a part of the support member is inserted into the insertion groove.
5. The communication device of claim 4, wherein the support member comprises: a support base; a protruding mount portion protruding from one side of the support base and configured to allow the elastic connecting member to be mounted thereon; protruding insertion portions formed on both sides of the protruding mount portion, and the gasket is disposed so that the protruding insertion portion is inserted into the insertion groove.
6. The communication device of claim 5, wherein the connector further comprises: a deformation gap formed between the support base and the elastic connecting member.
7. The communication device of claim 1, wherein the gasket is disposed to intersect the connector.
8. The communication device of claim 1, wherein the gasket is made of aluminum.
9. The communication device of claim 3, wherein a height of the insertion groove is greater than or equal to a half of a height of the body portion.
10. The communication device of claim 5, wherein a height of the protruding insertion portion is the same as the height of the insertion groove.
11. The communication device of claim 4, wherein the elastic connecting member comprises: a plate-shaped base, a first side wall formed on one side of the plate-shaped base; a first contact portion formed on one side of the first side wall; a second side wall formed on the other side of the plate-shaped base; and a second contact portion formed on one side of the second side wall, and the gasket is disposed so as to cover at least a part of the plate-shaped base.
12. The communication device of claim 4, wherein the support member is made of polycarbonate, Lusep, or Teflon.
13. The communication device of claim 1, wherein the instrument box comprises a plurality of radiating fins formed at least in a part thereof.
14. The communication device of claim 1, wherein the connector comprises: a support member disposed in the connection groove and contacting at least a part of the instrument box; and a conductive elastic connecting member disposed between the support member and the plurality of printed circuit boards, wherein the support member and the elastic connecting member are integrally formed.
15. The communication device of claim 1, wherein an upper surface and a lower surface of the gasket are formed in a flat shape.
16. The communication device of claim 1, wherein one surface of the gasket is formed to have a step.

17. An RF connector, comprising: an elastic connecting member, at least a part of which is elastically deformable, configured to electrically connect the printed circuit boards; a support member for supporting the elastic connecting member; and a gasket disposed to cover at least a part of the elastic connecting member.

18. The RF connector of claim 17, the support member comprises: a support base; a protruding mount portion protruding from one side of the support base and configured to allow the elastic connecting member to be mounted thereon; and protruding insertion portions formed on both sides of the protruding mount portion, and the gasket is disposed so that the protruding insertion portion is inserted into an insertion groove.

19. The RF connector of claim 17, wherein the gasket is disposed to intersect with the elastic connecting member.

20. The RF connector of claim 17, wherein the elastic connecting member comprises: a plate-shaped base, a first side wall formed on one side of the plate-shaped base; a first contact portion formed on one side of the first side wall; a second side wall formed on the other side of the plate-shaped base; and a second contact portion formed on one side of the second side wall, and the gasket is disposed so as to cover at least a part of the plate-shaped base.
